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SUBJECTIVE WORKSHEET

Acids, Bases and Salts

Science — Chemistry · Class 7 · Max Marks: 34 · Time: 1 hour

Section A — Very Short Answer (1 × 5 = 5)

1. Name the acid present in curd. [1]

2. Name the natural indicator obtained from lichen. [1]

3. Name the two products always formed when an acid reacts with a base. [1]

4. Name the base present in milk of magnesia. [1]

5. State the colour of turmeric indicator in a basic solution. [1]

Section B — Short Answer (2 × 5 = 10)

1. Define neutralisation and write its general word equation. [2]

2. Explain how an antacid relieves indigestion caused by too much acid in the stomach. [3]

3. A field is found to be too acidic for crops. State what should be added to it and give the reason. [2]

4. Describe how you would use turmeric paper to find out whether a given solution is basic. [2]

5. Give any two differences between acids and bases. [2]

Section C — Long Answer (5 × 3 = 15)

1. Describe an activity using red and blue litmus to classify three unknown solutions as acidic, basic or neutral. [5]

2. Explain, with three examples, how neutralisation is useful in everyday life. [5]

3. What are natural indicators? Name any three and state the colour each shows with an acid and with a base. [5]

Section D — Higher-Order Thinking (Give Reasons) (2 × 2 = 4)

1. A liquid changes the colour of neither red nor blue litmus. What can you conclude about it? Give one example. [2]

2. Why is a mild base such as baking soda, and not a strong base like sodium hydroxide, used on an ant's sting? [2]

Answer Key

Very Short Answer

1. Lactic acid.
2. Litmus.
3. A salt and water.
4. Magnesium hydroxide.
5. Red.

Short Answer

1. Neutralisation is the reaction in which an acid and a base react to cancel each other's effect: acid + base → salt + water (heat is released).
2. The stomach makes hydrochloric acid; when there is too much, it causes indigestion. An antacid is a mild base that neutralises the excess acid and relieves the discomfort.
3. Lime (a base, e.g. slaked lime/quick lime) should be added to neutralise the excess acid and make the soil suitable for crops.
4. Put a drop of the solution on turmeric paper. If it turns red, the solution is basic; if there is no change it is acidic or neutral.
5. Acids taste sour and turn blue litmus red; bases taste bitter, feel soapy and turn red litmus blue.

Long Answer

1. Put each solution on both red and blue litmus. If blue turns red → acidic; if red turns blue → basic; if neither changes → neutral. Record the results for the three solutions in a table and classify each.
2. Indigestion — antacid (base) neutralises excess stomach acid. Ant/bee sting — baking soda neutralises the acid. Soil treatment — lime neutralises acidic soil. Factory waste — basic substances neutralise acidic effluent.
3. Natural indicators are substances from nature that show different colours in acids and bases. Litmus: red in acid, blue in base. Turmeric: yellow in acid (no change), red in base. China rose: dark pink in acid, green in base.

Higher-Order Thinking (Give Reasons)

1. It is neutral (neither acidic nor basic), e.g. distilled water, common salt solution or sugar solution.
2. A strong base is corrosive and would damage the skin; a mild base neutralises the acid safely.